

Drone-based RGBT tiny person detection

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ABSTRACT

RGBT person detection benefits numerous vital applications like surveillance, search, and rescue. Meanwhile, drones can capture images holding broad perspectives and large searching regions per frame, which can notably improve the efficacy of large-scale search and rescue missions. In this work, we leverage the advantages of drone-based vision for RGBT person detection. The drone-based RGBT person detection task brings interesting challenges to existing cross-modality object detectors, e.g., tiny sizes of objects, modality-space imbalance, and position shifts. Observing that there is a lack of data and customized detectors for drone-based RGBT person detection, we contribute two new datasets and design a novel detector. The data contribution is two-fold. For one, we construct the first large-scale drone-based RGBT person detection benchmark RGBTDronePerson, which contains 6,125 pairs of RGBT images and 70,880 instances. Images are captured in various scenes and under various illumination and weather conditions. For another, we annotate the VTUAV tracking dataset and obtain its object detection version, named VTUAV-det. To tackle the challenges raised by this task, we propose a Quality-aware RGBT Fusion Detector (QFDet). Firstly, we design a Quality-aware Learning Strategy (QLS) to provide sufficient supervision for tiny objects while focusing on high-quality samples, in which a Quality-Aware Factor (QAF) is designed to measure the quality. Moreover, a Quality-aware Cross-modality Enhancement module (QCE) is proposed to predict a QAF map for each modality, which not only indicates the reliability of each modality but also highlights regions where objects are more likely to appear. Our QFDet remarkably boosts the detection performance over tiny and small objects, surpassing the strong baseline on mAP_{50}^{tiny} by 6.57 points on RGBTDronePerson and mAP_s by 3.80 points on VTUAV-det. The datasets, codes, and pre-trained models are available at <https://nnnnerd.github.io/RGBTDronePerson/>.

1. Introduction

Person detection (Dollar et al., 2012; Akshatha et al., 2023) is a crucial technology with wide applications in security, surveillance, crowd control, and search & rescue (Naidoo et al., 2011; Mukherjee et al., 2022). Person detection has achieved significant progress in the deep learning era. In particular, the recently emerging multispectral person detection techniques make up for many scenarios that fail to be tackled by RGB-based techniques. For example, the introduction of thermal modality alleviates the problem of poor performance under severe illumination conditions in RGB-based methods.

Simultaneously leveraging RGB and Thermal (RGBT) images for person detection integrates the complementary advantages of both modalities. More precisely, visible (RGB) images can provide abundant color and texture information when plenty of light is offered

(e.g., daytime), while thermal images are insensitive to illumination variations. Reversely, objects with similar heat radiation cannot be distinguished in thermal images, while visible images are not affected by the heat problem. In light of the complementarity of RGB and thermal data, several benchmarks for pedestrian detection (Hwang et al., 2015; González et al., 2016; Teledyne, 2018; Jia et al., 2021) have been proposed.

The drone-based images capture scenes at a long distance and provide a bird's eye view in a broad range (Zhu et al., 2021b; Benjamin et al., 2018; Nicolas et al., 2017). Drones thus possess the merit of spatial flexibility. RGBT data, as discussed above, provides around-the-clock useful information and possesses the merit of time flexibility. Therefore, the combination of RGBT data and drone-based images creates time-spatial flexibility, which is desperately needed in search & rescue. Unfortunately, there is no drone-based RGBT person detection

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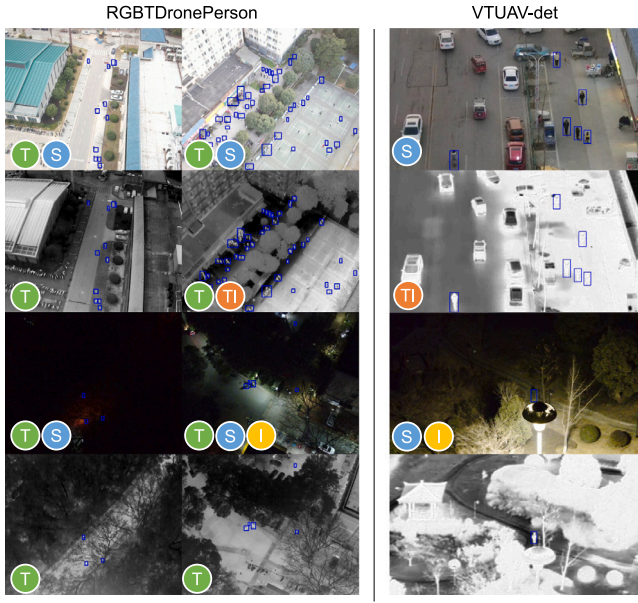


Fig. 1. Examples of drone-based person detection data. RGBTDronePerson and VTUAV-det are two datasets collected in this paper. Icons represent different challenges: “T” denotes **tiny object**; “TI” denotes **thermal imbalance**; “I” denotes **illumination imbalance**; “S” denotes **position shift**.

benchmark currently, hindering the related research and applications. To this end, this paper proposes a large-scale drone-based RGBT person detection benchmark, coming with two datasets (see Fig. 1). The first dataset is newly collected by drone-based RGBT cameras from various scenarios, named “RGBTDronePerson”, which contains 6125 pairs of RGBT images and 70,880 instances. The second dataset is based on the large-scale visible-thermal UAV tracking benchmark, VTUAV (Zhang et al., 2022), we re-annotate it and obtain its detection version called “VTUAV-det”, containing a total of 11,392 pairs of training images and 5378 pairs of testing image.

The drone-based RGBT person detection task comprises challenges both in aerial detection and multi-modal fusion, e.g., extremely small object sizes, low SNR, and complex complementary information fusion. The RGBTDronePerson presents an extremely small object size distribution with 98% of the instances being tiny objects (sizes smaller than 20 pixels), as shown in Fig. 2(a). Notably, the combination of these two fields imposes new challenges. The first is the modality-space imbalance problem. The drone-captured images cover a wide range of areas, resulting in illumination imbalance and thermal imbalance issues in one image, as shown in Fig. 2(b). The quality of the RGBT image pair is not only modality imbalanced, but also spatially imbalanced. While previous works tend to handle the imbalance of the whole modality or image, being agnostic of the region’s illumination and thermal conditions. Specifically, some illumination-aware methods (Li et al., 2019; Zhou et al., 2020; Zhang et al., 2023) predict an illumination value for the entire visible image. They commonly cannot identify which region is of higher quality in a certain image. Secondly, there are prominent position shifts between object pairs in different modalities. Most multispectral fusion methods assume that image pairs are geometrically aligned. Nevertheless, even registered RGBT image pairs are weakly aligned in practice as explained in the previous work (Zhang et al., 2019). Let alone in the drone-based RGBT imagery where there are many tiny objects, a slight deviation of the annotated bounding box will result in the box’s drift from the objects’ main body (Xu et al., 2022a). Therefore, the position shift is an unignorable issue in the drone-based RGBT person detection, which will lead to severe feature map mismatch and bounding box confusion. Fig. 2(c) shows some example image pairs

in RGBTDronePerson and illustrates the phenomenon of prominent objects’ position shifts.

To tackle these problems, especially the challenging tiny object detection and modality-space imbalance issues, we design a novel Quality-aware RGBT Fusion Detector (QFDet). The QFDet is mainly composed of two parts, namely a Quality-aware Learning Strategy (QLS) and a Quality-aware Cross-modality Enhancement (QCE) module. Concretely, the Quality-aware Learning Strategy (QLS) enhances the label assignment for tiny objects via a Scale-Invariant Wasserstein Distance (SIWD) and a Quality-Aware Factor (QAF). Previous studies improve the label assignment of tiny objects by designing location deviation-tolerant metrics (e.g. the Normalized Wasserstein Distance (NWD) (Xu et al., 2022a), the revised Bhattacharyya distance (Tang et al., 2022)), or seek balanced learning via hierarchical ranking strategies (Xu et al., 2022b). However, they only consider the location quality and are unaware of the classification confidence. Differently, QLS simultaneously leverages the scale-invariant localization quality SIWD and classification quality with QAF, obtaining high-quality label assignments for tiny objects. Moreover, in QLS we design a label assignment method that consistently exploits the prior and posterior knowledge to select high-quality samples for objects of different sizes. The Quality-aware Cross-modality Enhancement (QCE) not only indicates the reliability of each modality but also guides the detector to focus on regions where objects are more likely to appear. Compared to existing works (Li et al., 2019; Zhou et al., 2020; Zhang et al., 2023) that reweight the whole modality or the entire image, the QCE works more finely. We pose a modality-aware and region-aware soft weight to the feature map via the predicted QAF map, eliminating the effects of low-quality regions under illumination and thermal imbalance conditions.

In summary, the contributions of this paper are listed as follows:

- We construct a large-scale drone-based RGBT person detection benchmark RGBTDronePerson, which is the first to qualify spatial-time flexibility needed in search & rescue. The benchmark poses great challenges to existing algorithms. Additionally, we also build another drone-based RGBT person detection dataset VTUAV-det by labeling the VTUAV tracking dataset.
- We propose a Quality-aware RGBT Fusion Detector (QFDet), which comprises QLS and QCE. QLS simultaneously leverages localization & classification and combines the strengths of prior & posterior knowledge, obtaining high-quality samples. QCE achieves complementary fusion by accessing cross-modality quality in a prediction-guided way.
- Extensive experiments on RGBTDronePerson and VTUAV-det datasets are conducted to validate the effectiveness of the proposed method. Our QFDet remarkably boosts the detection performance over tiny and small objects, surpassing the strong baseline on mAP_{50}^{tiny} by 6.57 points on RGBTDronePerson and mAP_s by 3.80 points on VTUAV-det.

The rest of this paper is organized as follows: In Section 2, we briefly describe the related works. In Section 3, we introduce the proposed RGBTDronePerson dataset. Section 4 gives a detailed description of QFDet. Then, we validate the effectiveness of the proposed method and discuss the experimental results in Section 5. Finally, we conclude this paper in Section 6.

2. Related work

RGBT detection datasets. RGBT data is widely used in multispectral person detection. KAIST (Hwang et al., 2015), CVC-14 (González et al., 2016), FLIR ADAS (Teledyne, 2018) are three classic datasets in multispectral pedestrian detection, all of which are vehicle-viewed and are studied for autonomous driving. Recent years have seen wider studies on multispectral fusion. More datasets are proposed: RGBT-CC (Liu et al., 2021) and DroneRGBT (Peng et al., 2021) for crowd counting,

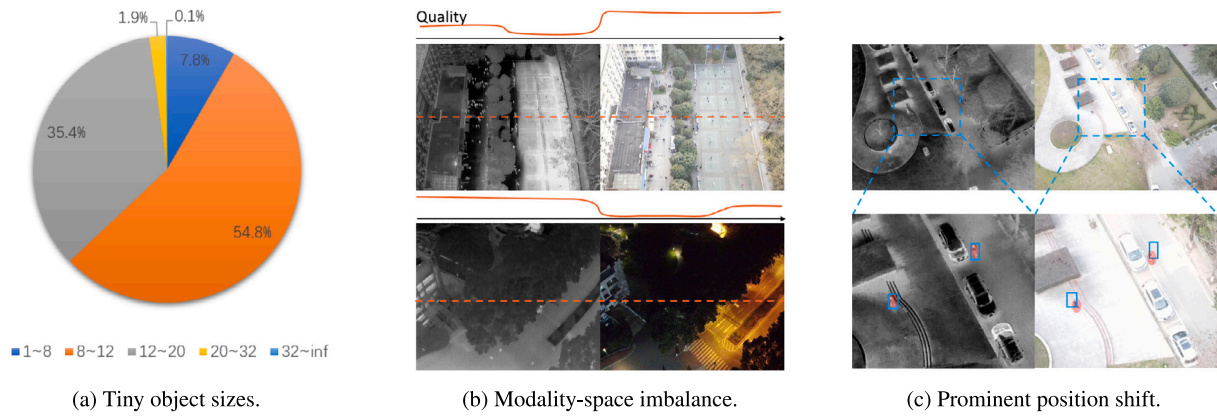


Fig. 2. Challenges put forward by RGBTDronePerson. (a) The object size distribution shows that the majority of RGBTDronePerson consists of tiny objects (sizes smaller than 20). (b) The image quality for person detection varies along the reference line. The quality curves are drawn intuitively for a straightforward demonstration. In the upper left thermal image, the ground on the right side absorbs heat, reaching similar thermal radiation with human objects. Therefore, persons are concealed in this area. In the bottom right visible image, persons are invisible in the dark area. (c) The bounding boxes of thermal images shift away from objects in visible images.

DroneVehicle (Sun et al., 2022) for vehicle detection, LLVIP (Jia et al., 2021) for image fusion and pedestrian detection. DroneVehicle builds a bridge between drone-based detection and multispectral detection. Different from most multispectral pedestrian detection datasets, the proposed RGBTDronePerson is captured from a drone and has various scenes from different viewpoints. Different from DroneVehicle, RGBTDronePerson is a person-centric detection dataset and objects have extremely small sizes. TinyPerson (Yu et al., 2020a) also presents persons captured at a long distance and in a very low resolution. However, TinyPerson only has a visible modality and is not applicable in the nighttime. The attributes of drone-based and RGBT data in various scenes make RGBTDronePerson generalize to much more scenes in search & rescue.

RGBT fusion-based object detection. RGBT fusion is a subproblem of multi-modal fusion. In multi-modal fusion, there are three key factors to consider, namely what to fuse, how to fuse, and when to fuse (Feng et al., 2021). As for how to fuse, a basic and simple fusion mechanism is element-wise operations on modality feature maps, i.e., addition, average, max. Another commonly used method is concatenation with a 1×1 convolution. Efforts have been made on how to fuse information from each modality adaptively. Some (Li et al., 2019; Guan et al., 2019; Zhou et al., 2020; Zhang et al., 2023) design an illumination-aware subnetwork to predict weights so that at night visible features will be suppressed. Some (Zhang et al., 2019; Kim et al., 2021b) calculate uncertainty to decide which modality is more reliable. As for when to fuse, in previous work (Liu et al., 2016; Li et al., 2019), different fusion structures are studied based on Faster R-CNN (Ren et al., 2017). Depending on when to fuse in the network, fusion structures can be divided into input fusion, early fusion, middle fusion, late fusion, and score fusion. Experiments (Li et al., 2019; Wolpert et al., 2020) have shown that middle fusion can achieve better detection performance. Chen et al. (Chen et al., 2022b) designed a Bayesian-based probabilistic ensembling method to fuse the detection results of multi-modal models, which is a non-learned late-fusion method. Numerous work (Zhou et al., 2020; Zhang et al., 2020c, 2021; Kim et al., 2021b,a; Zhang et al., 2023) based on the two-stream structure has been done to explore better fusion structure. Recent work (Liu et al., 2021; Zhang et al., 2022) innovates in fusion structure, instead of two-stream, adopting a three-branch structure.

Tiny object detection. It is a challenging task to detect tiny objects since they contain very few pixels. Previous work has been done in multi-scale feature learning, feature enhancement, data augmentation, label assignment, etc. In multi-scale feature learning, Feature Pyramid Network (FPN) (Lin et al., 2017a) presents a top-down architecture with lateral connections to supplement high-level semantic information in low-level feature maps. Other attempts have been made to

improve FPN, such as PANet (Liu et al., 2018), BiFPN (Tan et al., 2020), and Recursive FPN (Qiao et al., 2021). Compared to FPN-based methods, some works directly construct customized multi-scale features for the aerial imagery (Deng et al., 2018; Zheng et al., 2020; Cheng and Han, 2016). Besides the multi-scale feature learning methods, some works propose to enhance the feature representation for tiny objects. In DRENet (Chen et al., 2022a), an auxiliary network branch, i.e., degraded reconstruction enhancer is proposed to help the backbone focus on tiny objects instead of the vast background. Other works, PGAN (Li et al., 2017), MTGAN (Bai et al., 2018), etc., propose super-resolution-based methods to enhance the feature representations. In addition, augmenting tiny object data is an easy and effective way to improve tiny object detection performance (Bosquet et al., 2023). Recent advances in label assignment reinvigorate the research of tiny object detection, NWD-RKA (Xu et al., 2022a) and RFLA (Xu et al., 2022b) are proposed to boost the tiny object detection via balanced learning and location deviation-tolerant metrics. Nevertheless, these label assignment methods only consider location quality. In this paper, we simultaneously consider location and classification quality with a Quality-aware Learning Strategy (QLS). In QLS, we design a scale-invariant location metric SIWD and further a prediction-aware QAF, obtaining high-quality learning for tiny objects. Moreover, we design a label assignment method that consistently exploits the prior and posterior knowledge to select high-quality samples. From the aspect of data, several aerial datasets (Wang et al., 2021b; Yu et al., 2020b; Xu et al., 2022a) have been proposed to promote the study of tiny object detection. However, these datasets only have a single modality and are limited in the daytime. In this paper, we construct an RGBT dataset for tiny person detection, which further widens the scope of application for tiny object detection.

3. RGBTDronePerson benchmark

3.1. Description

Data collection. Image pairs in RGBTDronePerson are captured in a dual camera-equipped drone, DJI Mavic 2 Enterprise Advanced. We use it to record dual-spectral videos on the fly and obtain 1920×1080 visible videos and 640×512 thermal videos. We sample image pairs from the videos and then crop and resize the visible images to 640×512 . No post-processing is applied to register or align image pairs. Considering different lighting conditions, we collect abundant images both in the daytime and nighttime. And considering different temperature conditions, the data collection work spans from winter to summer in Wuhan, China. The drone flies at a height of 50m–80 m and shoots at an angle of 45° – 60° (a small number of images are shot from a vertical angle).

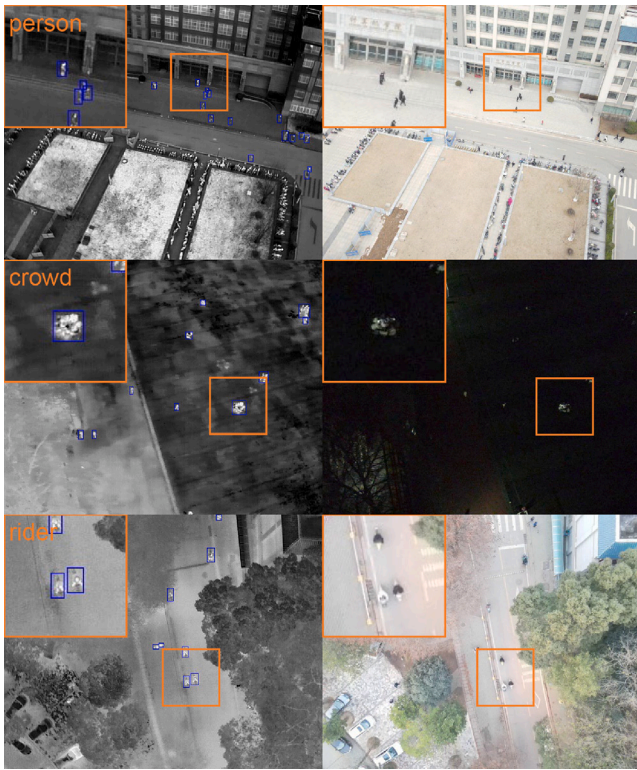


Fig. 3. Instance examples and corresponding image pairs.

The dataset contains 6125 pairs of RGBT images. We randomly split the dataset into training and validation sets. The training set consists of 4900 pairs and the validation set consists of 1225 pairs.

Annotation. We use LabelImg to label thermal images. It should be noted that when labeling thermal images, the labeler also looks into its visible pair. In this way, although we only annotate thermal modality, information from visible modality is exploited to a certain degree, and objects that are hard to be seen in thermal modality can be annotated when they are evident in visible modality. We assign 4 possible labels to objects, which are “person”, “rider”, “crowd” and “uncertain”. A single person is labeled as “person”. A person riding a scooter or bicycle is labeled as “rider”. When more than two persons are crowded together and cannot be told apart, we label it as “crowd”. When an object is not certain to be a person for the labeler, it is labeled as “uncertain”. We obtain 70,880 annotated instances after double check. Instance examples are shown in Fig. 3. In the training set, there are 43,006 instances of “person”, 4869 instances of “rider”, and 7316 instances of “crowd”.

Evaluation. Following TinyPerson (Yu et al., 2020b), we use average precision (AP) for performance evaluation. IoU thresholds 0.25 and 0.50 are chosen to determine true positive predictions. Due to the tiny sizes of objects, a small position shift between a detection box and a GT will result in a dramatic IoU drop. Thus, IoU threshold 0.25 is used here. The category “uncertain”, which accounts for 2% of the total annotations, is not considered during the training stage and evaluation process.

Properties. Compared with KAIST and LLVIP, the object sizes in RGBTDronePerson are extremely small. Following the definition in TinyPerson, object sizes that are smaller than 20 pixels are defined as “tiny”, between 20–32 pixels are defined as “small”, and bigger than 32 pixels are defined as “reasonable”. In the training set, 98.0% of objects are tiny objects. And in the whole dataset, there are 54,178 instances of “person”, 6080 instances of “rider”, and 9176 instances of “crowd”. Table 1 and Fig. 4 reports the comparison with other related object detection datasets.

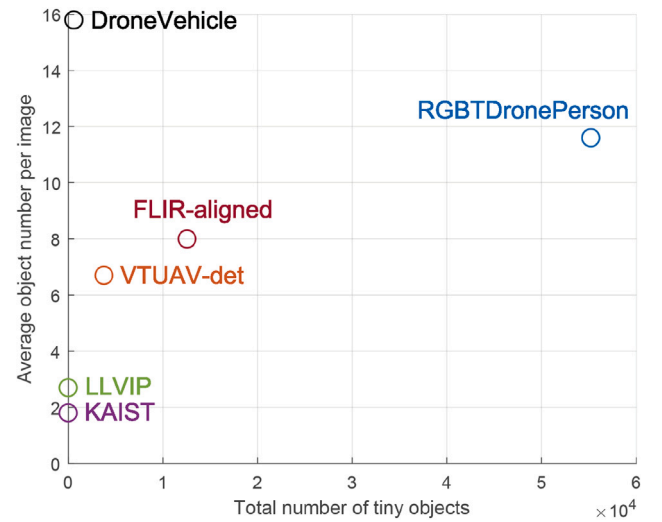


Fig. 4. Comparison with other multispectral object detection datasets. The radius of each circle demonstrates the total number of tiny objects.

3.2. Challenges

Tiny sizes of objects. In RGBTDronePerson, the average object size is only 11.7 pixels, and 98.0% of objects are smaller than 20 pixels. Because tiny objects contain little information and their ground truth boxes have high noise, it is hard for common detectors to perform well on tiny object detection. RGBTDronePerson is the first RGBT dataset for tiny person detection. The tiny size objects in thermal modality not only introduce the problem of few discriminative features but also bring in the problem of low contrast. In thermal modality, objects captured at a long distance are easily lost in the noise and unwanted waves. It is very challenging to detect tiny objects in both visible and thermal modalities.

Modality-space imbalance. Visible and thermal sensors perform differently under different conditions. An apparent example is that at night visible modality is less reliable than the thermal modality because of insufficient illumination. However, the quality imbalance problem in drone-viewed images not only exists between the modalities but also exists between different regions. The drone-captured images cover a wide range of areas, resulting in illumination imbalance and thermal imbalance in one specific image. For example, due to the presence of artificial light at night, the illumination condition varies across different regions in one visible image. Likewise, the drone-captured image covers different types of ground, thus, the background radiation is also different in one thermal image. Existing algorithms (Li et al., 2019; Zhou et al., 2020; Zhang et al., 2023) only predict a value for the entire image/modality to present the illumination condition or reliability of the image pair. However, it is not exact to present reliability or quality of modality in this way when there is also space imbalance, this could lead to detectors’ over-optimization on low-quality regions in the training process.

Prominent position shifts between modalities. Originated from the differences in imaging characteristics between visible and thermal sensors, visible and thermal images are not strictly aligned. Moreover, visible and thermal sensors cannot be exactly in the same place on hardware in the real world. These two reasons lead to the position shift of objects between modalities. Especially in RGBTDronePerson, due to the extremely small sizes of objects, small position shift on the whole image is prominent to objects. This could lead to severe feature map mismatch and bounding box confusion. Some multispectral object detection algorithms considering the misalignment problem between modalities require annotations for each modality, which is

Table 1

Comparison with other related object detection datasets.

Dataset	Modality	View	Person-centric	Absolute size	Relative size	Resolution	Year
KAIST (Hwang et al., 2015)	RGBT	Vehicle	✓	52.1 ± 7.2	0.091 ± 0.302	640×512	2015
FLIR-aligned (Zhang et al., 2020c)	RGBT	Vehicle	✓	40.4 ± 6.4	0.705 ± 0.266	640×512	2020
LLVIP (Jia et al., 2021)	RGBT	Surveillance	✓	142.3 ± 11.9	0.124 ± 0.353	1080×720	2021
TinyPerson (Yu et al., 2020b)	RGB	Free	✓	18.0 ± 17.4	0.012 ± 0.010	–	2020
DroneVehicle (Sun et al., 2022)	RGBT	Drone	✗	49.3 ± 7.0	0.064 ± 0.253	840×712	2022
VTUAV-det	RGBT	Drone	✓	72.9 ± 8.5	0.051 ± 0.225	1920×1080	2023
RGBTDronePerson	RGBT	Drone	✓	11.7 ± 3.4	0.020 ± 0.143	640×512	2023

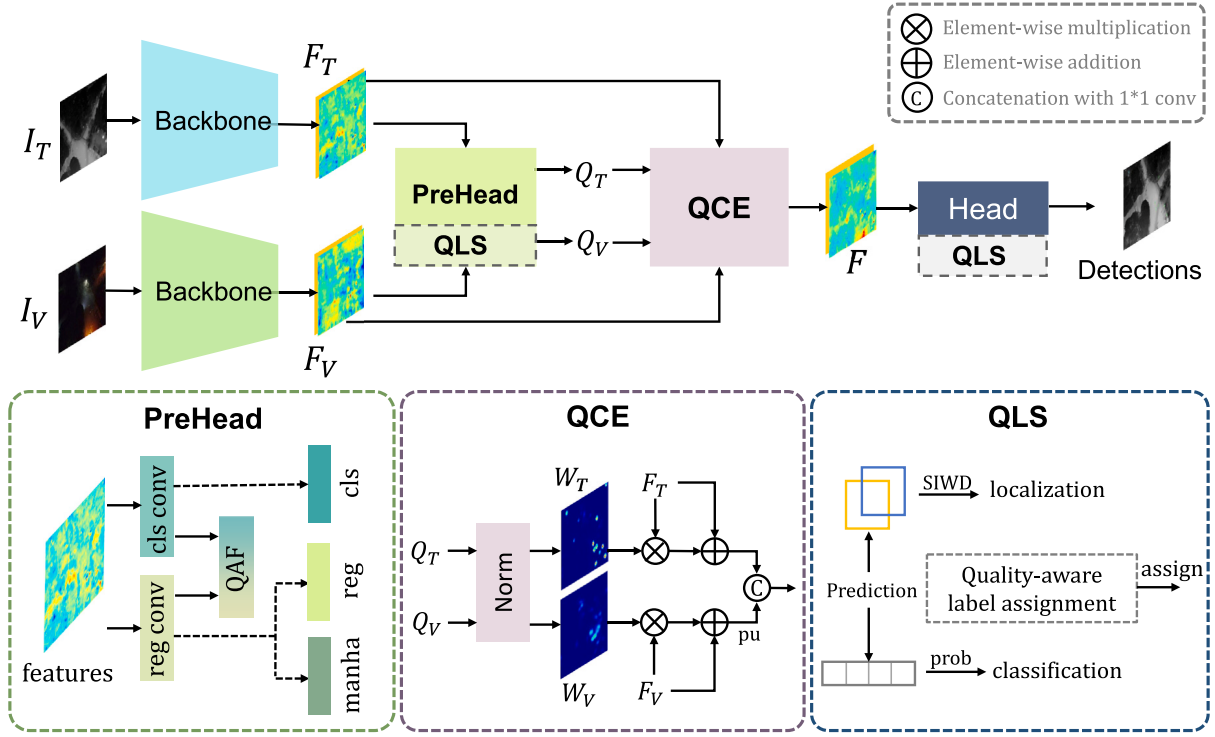


Fig. 5. The structure of QFDet. In the detection head, we propose QLS to strengthen the training for tiny objects. The final detection head takes the fused features as input. The thermal and visible features are fused in a prediction-guided way. Each modality feature will first be fed into PreHead to predict QAF maps. Q_T and Q_V in the figure denote QAF maps for thermal and visible modalities, respectively. Then the multi-modal features are fused adaptively in the QCE module. “pu” in the QCE module denotes the pool-upsample operation.

labor-intensive. In RGBTDronePerson, only the thermal modality is aligned with the annotations. It is necessary and challenging to study how to alleviate position shifts between modalities with minimum labor.

4. Quality-aware RGBT fusion detector

4.1. Overview

The structure of the proposed QFDet is shown in Fig. 5. QFDet addresses the imbalance problem by simultaneously leveraging the localization and classification quality for each modality. In the PreHead, QAF maps will be predicted for each modality, which not only indicate the reliability of different modalities but also imply where is possible to be an object. Given the QAF maps, the thermal and visible features are enhanced and fused accordingly. In the PreHead and the detection head, QLS is designed to improve the detection performance on tiny objects as well as objects of various sizes. We will elaborate on QLS first in the next subsection. Then, PreHead, QCE, and the loss of QFDet will be introduced.

4.2. Quality-aware learning strategy

It is challenging to assign samples of high quality to tiny GTs. In many state-of-the-art generic object detectors (Zhang et al., 2020b; Ren

et al., 2017; Lin et al., 2017b; Cai and Vas., 2018), IoU is used for measuring the similarity between two bounding boxes. While previous works have pointed out two main drawbacks of IoU for tiny objects. The first one is the drastic drop of IoU value to the slight box deviation (Xu et al., 2021) and the second is its incapability to measure non-overlapping boxes (Wang et al., 2021a; Xu et al., 2022b), leading to insufficient and low-quality training samples for tiny objects.

Some works successfully alleviate these issues by proposing new metrics like the NWD (Wang et al., 2021a) and seeking balanced learning like the RKA (Xu et al., 2022a) and RFLA (Xu et al., 2022b). These works have pushed forward the label assignment of tiny objects by a large margin. However, we still observe two remaining issues for tiny object detection label assignment. On the one hand, the NWD metric lacks scale-invariance since it is based on the scale-variant Wasserstein distance (Olkin and Pukelsheim, 1982; Yang et al., 2021). On the other hand, existing assignment strategies for tiny object detection work in a prediction-agnostic manner, which means that they cannot dynamically determine high-quality samples in the training process. Towards this end, we simultaneously consider localization and predicted confidence with a Quality-aware Learning Strategy (QLS). In QLS, we design a scale-invariant localization metric SIWD, and further, a prediction-aware QAF, obtaining high-quality learning for tiny objects. Moreover, we design a label assignment method (QLA) considering prior and posterior knowledge to further select high-quality positive samples.

Additionally, we notice that the centerness (Tian et al., 2019a) also encounters the drastic drop issue when the predicted box slightly deviates from the tiny GT box, suppressing correct predictions in NMS. We propose a new metric based on the Manhattan distance to tackle this issue, called manhattaness.

SIWD. To begin with, we have a brief review of the NWD (Xu et al., 2022a), it first models horizontal bounding boxes (c_x, c_y, w, h) as 2-D Gaussian distributions $\mathcal{N}(\mu, \Sigma)$:

$$\mu = \begin{bmatrix} c_x \\ c_y \end{bmatrix}, \Sigma = \begin{bmatrix} \frac{w^2}{4} & 0 \\ 0 & \frac{h^2}{4} \end{bmatrix} \quad (1)$$

Further, Wasserstein distance is used to measure the distance between two distributions. For two distributions $\mathcal{N}_1, \mathcal{N}_2$, the second-order Wasserstein distance is given by:

$$W_2^2(\mathcal{N}_1, \mathcal{N}_2) = \|\mu_1 - \mu_2\|_2^2 + \|\Sigma_1^{1/2} - \Sigma_2^{1/2}\|_F^2 \quad (2)$$

Normalizing $W_2^2(\mathcal{N}_1, \mathcal{N}_2)$ to (0, 1] yields NWD:

$$\text{NWD}(\mathcal{N}_1, \mathcal{N}_2) = \exp\left(-\frac{\sqrt{W_2^2(\Sigma_1, \Sigma_2)}}{C}\right) \quad (3)$$

where C is set to the average object scale of the training set (Xu et al., 2022a).

The Wasserstein distance is not scale-invariant, i.e., if the mean vectors μ_1, μ_2 and the square root of variation matrices $\Sigma_1^{1/2}, \Sigma_2^{1/2}$ are multiplied with a full-rank scale factor $K = kI$ (I denotes the identity matrix), the Wasserstein distance between the two distributions will change. This property makes NWD not scale-invariant, which indicates that it may not work very well in the scenario where objects have large-scale variation.

To tackle this problem, we design a Scale-Invariant Wasserstein-based Distance (SIWD). We first obtain a scale matrix:

$$S = \Sigma_1^{1/2} + \Sigma_2^{1/2} = \begin{bmatrix} \frac{w_1 + w_2}{2} & 0 \\ 0 & \frac{h_1 + h_2}{2} \end{bmatrix} \quad (4)$$

where the diagonal elements are the arithmetic mean of the widths and heights of the two bounding boxes.

We scale the two distributions by multiplying the mean vectors μ_1, μ_2 and the square root of variation matrices $\Sigma_1^{1/2}, \Sigma_2^{1/2}$ with the inverse matrix of S . Then we calculate the Wasserstein distance between the scaled distributions, yielding $D^2(\mathcal{N}_1, \mathcal{N}_2)$:

$$D^2(\mathcal{N}_1, \mathcal{N}_2) = \|S^{-1}(\mu_1 - \mu_2)\|_2^2 + \|S^{-1}(\Sigma_1^{1/2} - \Sigma_2^{1/2})\|_F^2 \quad (5)$$

Further, normalizing $D^2(\mathcal{N}_1, \mathcal{N}_2)$, we obtain SIWD:

$$\text{SIWD}(\mathcal{N}_1, \mathcal{N}_2) = \exp(-2\sqrt{D^2(\mathcal{N}_1, \mathcal{N}_2)}) \quad (6)$$

As shown in Fig. 6, the properties of SIWD are investigated experimentally. Besides scale invariance, SIWD also overcomes the IoU's problem of drastic value drop and the disability of measuring non-overlapping boxes.

Quality-aware label assignment. Many state-of-the-art anchor-based detectors (Lin et al., 2017b; Zhang et al., 2020a,b) utilize the IoU between anchors and GT boxes to select positive samples. However, as analyzed in several works (Ming et al., 2021; Kim and Lee, 2020), this static heuristic fails to dynamically screen out low-quality or noisy samples. Several attempts have been made to consider the IoU between prediction boxes and GT boxes. Nevertheless, solely relying on the predictions for quality measurement will lead to instability at the early stages of the training process. In QLA, we propose to combine the strengths of anchors and predictions to select high-quality positive samples.

We call the location metric between anchors and GT boxes prior location score, the location metric between prediction boxes and GT boxes prediction location score for convenience. Every feature point has a pre-defined anchor and a prediction box, where we use SIWD

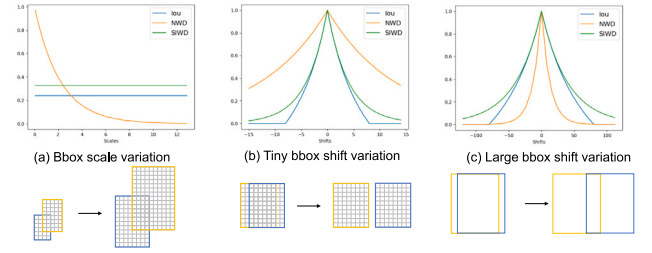


Fig. 6. Value variations of different metrics. (a) gives the value variations of IoU, NWD, and SIWD when the bounding box is scaled down and up. Both IoU and SIWD are scale-invariant. (b) shows that SIWD does not drop drastically when the bounding boxes are tiny (8×10 in this example); SIWD also gives sufficient discrimination between close and un-overlapped bounding boxes. (c) shows that SIWD does not drop drastically either when the bounding boxes are large (80×100 in this example).

to evaluate the location score between boxes. We calculate the prior location score and prediction location score for every feature point and choose the higher value as a Hybrid Location Quality (HLQ), which indicates the localization ability of the feature point. This can be formulated as:

$$\text{HLQ}(i, j) = \max(\text{SIWD}_a(i, j), \text{SIWD}_p(i, j)) \quad (7)$$

where i denotes the i th feature point, j denotes the j th GT, and subscript a, p denote anchor and prediction, respectively.

To alleviate the inconsistency between localization and classification, we simultaneously consider the classification and location scores generated by this feature point. Therefore, we define the QAF between feature point i and GT j as:

$$Q_{i,j} = \mathbb{1}_S(i) \cdot (\text{HLQ}(i, j)^\alpha \cdot \text{prob}(i, C_j)^{1-\alpha}) \quad (8)$$

where $\mathbb{1}_S$ denotes the spatial prior, $\text{prob} \in [0, 1]$ is the classification probabilities, C_j denotes the class of GT j , and hyperparameter $\alpha \in [0, 1]$ adjusts the ratio between localization and classification. The value of QAF ranges from 0 to 1. In this way, the designed QAF is not fully dependent on the network's responses but regulated by the ground truths. Specifically, the HLQ is based on the location score between the anchor/prediction and ground truths. If the classification response is incorrect, the obtained location score will be low, which will finally lead to a low QAF score, eliminating the negative effects brought to network learning.

The spatial prior in many state-of-the-art detectors, e.g., FCOS, and ATSS, is that the center of the anchor should fall into a GT. However, we discover that due to the extremely small size of tiny objects, many samples are filtered out by the spatial prior even though they have a notable SIWD value with the GT box. Under this circumstance, many tiny GT boxes fail to be assigned with any positive samples. In QLA, we consider not only the anchor center but also the prediction center. Concretely, if the center of the anchor or its corresponding prediction falls into a GT, we consider it to satisfy the spatial prior.

We choose top- k samples whose center distance to GT is the closest as candidates for each GT. After that, we calculate the mean and std value of candidate QAF and set it as the dynamic threshold. Samples with higher QAF are assigned as positives. Similar to ATSS (Zhang et al., 2020b), this threshold-based label assignment method adaptively assigns a number of positive samples to different GTs. Therefore, it can provide sufficient training for objects of different sizes. It should be noted that ATSS assigns positive samples merely based on the IoUs between pre-defined anchors and GTs, while we calculate the cost matrix based on the QAF values of feature points to assign positive samples. In this way, discriminative features for localization and classification can be better captured.

Manhattaness. As shown in Fig. 7(a), in tiny GT boxes, the centerness (Tian et al., 2019b) decays rapidly when the positive sample

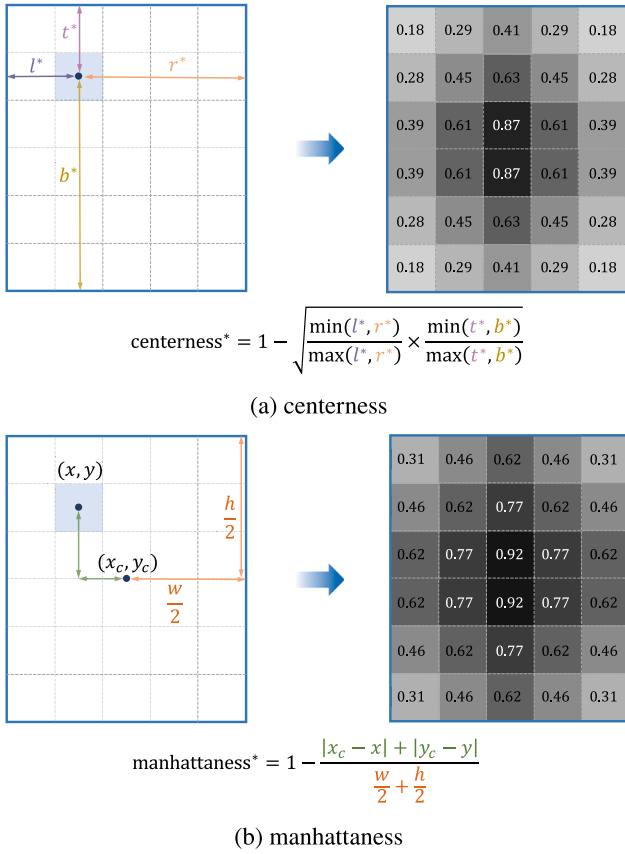


Fig. 7. Comparison between original centerness and the proposed manhattaness metric. Given a tiny GT box of size, 6×7 (the pixels occupied by four sides of the bounding box are omitted for convenience), centerness and manhattaness values for every pixel inside the box are shown on the right side.

moves away from the center. Given that the predicted confidence will be multiplied with a centerness score for a certain sample before NMS, the confidence of tiny objects will be significantly suppressed, making some true predictions falsely deleted. To tackle this, we propose a manhattaness metric to replace the centerness for tiny objects. We define center/corner as a pseudo point at the box's center/vertices. Further, we apply the Manhattan distance to measure the distance between pixels and pseudo points. The manhattaness value of a sample located at (x, y) is defined by:

$$\text{manhattaness}^* = 1 - \frac{|x_c - x| + |y_c - y|}{w/2 + h/2} \quad (9)$$

where (x_c, y_c) is the location of the box center, $|x_c - x| + |y_c - y|$ denotes the Manhattan distance between the pixel and the GT's center; $w/2 + h/2$ denotes the Manhattan distance between the center and the corner. Eq. (9) can also be formulated as:

$$\text{manhattaness}^* = \frac{\min(l^*, r^*) + \min(t^*, b^*)}{(l^* + r^* + t^* + b^*)/2} \quad (10)$$

where l^* , t^* , r^* , and b^* are the distances from the location to the four sides of the bounding box, as shown in Fig. 7(a).

4.3. Quality-aware cross-modality enhancement

As stated in Section 1, there is not only a modality imbalance but also a spatial imbalance in RGBT aerial image pairs. It is not fine enough to predict a single value to describe the reliability of the whole modality. To effectively fuse complementary information, for each feature point pair (corresponding points in two modalities), we propose to address the one with higher quality. Following the design

in the previous section, we use QAF to measure the quality, which is measured by considering the Hybrid Location Quality (HLQ) and the predicted classification confidence. Therefore, the complementary information is fused and utilized effectively.

Due to the large misalignment between the annotated box location and the real object's location in the visible modality, it is hard for the network to learn an informative representation of the visible modality. To alleviate this problem, we train a shared head to perform detection in both modalities before fusion and the final detection head. The abundantly shared edge, texture, and semantic information of the thermal and visible modalities makes it possible to have the shared head produce reasonable predictions on the visible modality, given the thermal modality and its aligned annotations. The shared head, named PreHead, not only learns to detect objects but also predicts QAF maps for both modalities.

The structure of the PreHead is similar to the detection head but has an additional QAF prediction subhead. Specifically, there are regression *conv* layers and classification *conv* layers in PreHead. Given backbone-extracted features as input, regression *conv* layers yield *reg* features, while classification *conv* layers yield *cls* features. Bbox regression is performed on *reg* features and classification is made on *cls* features. Differently, the quality prediction proceeds on the concatenation of *reg* features and *cls* features. The regression heads, classification, and manhattaness heads are excluded at inference time.

The label assignment and QAF definition in PreHead are the same as in the final prediction head. After label assignment, we obtain QAF map target Q^* for each modality:

$$Q_i^* = \max_j(Q_{i,j}) \quad (11)$$

where i denotes the i th feature point, and j denotes the j th GT. This means we take the max QAF value between feature point i and all GTs as the QAF of this point. For negative samples, the QAF target is set to zero.

Given the predicted QAF maps $Q_T \in R^{H \times W}$ and $Q_V \in R^{H \times W}$, we initially perform minmax normalization across these two maps. The process can be formulated as follows:

$$W_T = \frac{Q_T - \min(Q_T, Q_V)}{\max(Q_T, Q_V) - \min(Q_T, Q_V)} \quad (12)$$

$$W_V = \frac{Q_V - \min(Q_T, Q_V)}{\max(Q_T, Q_V) - \min(Q_T, Q_V)}$$

Then the visible and thermal features will be enhanced as described in the following formula:

$$\hat{F}_T = (1 + W_T) \odot F_T \quad (13)$$

$$\hat{F}_V = (1 + W_V) \odot F_V$$

where $\odot$ denotes element-wise multiplication, and F denotes the fused features.

Considering that the visible modality is misaligned with the thermal modality, we apply max-pooling on the visible features. Then we upsample the pooled feature and fuse it with the original feature. The procedure can be expressed by:

$$\tilde{F}_V = \text{Conv}_{1 \times 1}(\hat{F}_V \oplus \hat{F}_V^u) \quad (14)$$

where $\oplus$ denotes channel-wise concatenation, $\hat{F}_V^u$ denotes the pooled visible feature. In this way, contextual information is also enhanced in $\tilde{F}_V$.

Finally, the enhanced thermal and visible features are fused by a 1×1 convolution layer:

$$F = \text{Conv}_{1 \times 1}(\hat{F}_T \oplus \tilde{F}_V) \quad (15)$$

4.4. Loss

The total loss of QFDet consists of two parts, detection head loss, and PreHead loss. The losses can be given by:

$$\begin{aligned} L_{tot} &= L_{det} + L_{pre} \\ L_{det} &= L_{reg} + L_{cls} \\ L_{pre} &= L_{preg} + L_{pcl} + L_q \end{aligned} \quad (16)$$

where L_{det} is the detection head loss, L_{pre} is the PreHead loss. L_{reg} , L_{preg} denote bounding box regression loss in detection head and PreHead, respectively, and L_{cls} , L_{pcl} do likewise. The regression task is supervised by GIoU loss (Rezatofighi et al., 2019) and the classification task is supervised by Focal loss (Lin et al., 2017b). L_{preg} and L_{pcl} are given by:

$$\begin{aligned} L_{preg} &= 0.5 \times (\text{GIoUL}(b_t, b^*) + \text{GIoUL}(b_v, b^*)) \\ L_{pcl} &= 0.5 \times (\text{FL}(c_t, c^*) + \text{FL}(c_v, c^*)) \end{aligned} \quad (17)$$

where $\text{GIoUL}(\cdot)$ and $\text{FL}(\cdot)$ denote the GIoU loss and Focal loss, respectively. b and c denote regression and classification prediction, respectively. Subscript t and v denote thermal and visible modality, respectively. $*$ denotes the target.

The QAF loss L_q is measured by mean square error (MSE) loss:

$$L_q = 0.5 \times \text{MSE}(Q_T, Q_T^*) + 0.5 \times \text{MSE}(Q_V, Q_V^*) \quad (18)$$

where Q_T, Q_V denote QAF maps for thermal and visible modalities, respectively.

5. Experiment

5.1. Implementation detail

Training and testing details. We implement the model with the Pytorch (Paszke et al., 2019) and MMDetectionV2 framework (Chen et al., 2019). All image inputs are resized to 640×512 (keeping ratio). The network backbone is ResNet50 (He et al., 2016) with FPN unless stated otherwise and is pre-trained on ImageNet (Deng et al., 2009). The hyperparameter α is empirically set to 0.8. The model is trained for 12 epochs by an SGD optimizer with a momentum of 0.9, a learning rate of 0.01, a weight decay factor of 0.0001, and a batch size of 8. To balance the losses of different levels, we reweigh the detection loss as in Yang et al. (2022). For RGBTDronePerson, the re-balance weights are all set to 1. For VTUAV-det, the re-balance weights are set to linearly growing from 1 to 5. To avoid overfitting, we take the epoch with the best testing results as the final model. As for data augmentation, we use random flip. The IoU threshold for test time NMS is set to 0.3 for RGBTDronePerson; for VTUAV-det, it is 0.5. The confidence score threshold is set to 0.05. Experiments are carried out on a single TITAN X GPU. Besides, we implement a stronger model denoted by QFDet* that utilizes P2-P6 feature maps for detection (in contrast to its counterpart using P3-P7).

Datasets. Main experiments are performed on the proposed RGBT-DronePerson dataset. Apart from RGBTDronePerson, we have collected another drone-based person detection dataset, VTUAV-det. VTUAV (Zhang et al., 2022) is a large-scale benchmark for visible-thermal UAV tracking, containing a variety of scenes and weather conditions. In this work, we collect sequences that contain persons and sample one image out of every 50 frames to form a drone-based person detection dataset. The train/test split follows the original VTUAV dataset. The annotation strategy is the same as that of RGBTDronePerson. We only consider the “person” category in this dataset. There are a total of 11,392 pairs of training images and 5378 pairs of testing images. In the training set, 74,010 instances of persons are annotated. Unlike RGBTDronePerson, which mainly consists of tiny objects, VTUAV-det presents more significant scale variation. Therefore, we take the mean Average Precision (mAP) used in COCO (Lin et al., 2014) evaluation

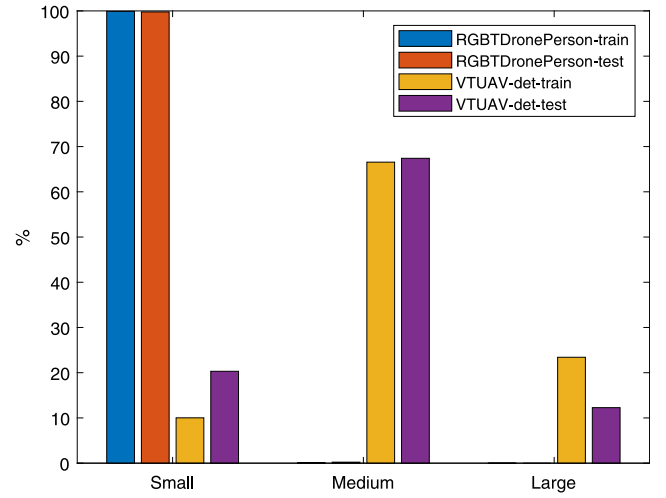


Fig. 8. Size distribution of VTUAV-det and RGBTDronePerson datasets under COCO definition.

as the evaluation metric. According to the COCO evaluation protocol, object sizes are categorized into “small”, “medium”, and “large”. The VTUAV-det training set has 7651 small objects, 50,794 medium objects, and 17,869 large objects. The testing set has 9864 small objects, 32,731 medium objects, and 5960 large objects. We show the size distribution of VTUAV-det and RGBTDronePerson datasets under COCO definition in Fig. 8. Different from RGBTDronePerson, the view in VTUAV-det is more diverse, resulting in greater scale variation. Moreover, the thermal sensor of VTUAV-det is different and the thermal image exhibits differences compared with thermal images in RGBTDronePerson. Examples of VTUAV-det images are shown in Fig. 9.² The diversity ensures that detection results on these two datasets can well reflect the method’s generalization ability.

Baseline detectors customized for RGBT object detection. For comparison, we implement representative state-of-the-art object detectors for RGBT detection as baseline detectors. These baselines adopt two-stream structures where two backbones extract visible and thermal features, respectively. Multiple levels of features from two branches are then fused by concatenation with 1×1 convolution. Noticing that ATSS shows satisfying performance in terms of accuracy and speed on tiny objects, we choose ATSS as the strong baseline for our method.

5.2. Comparison with state-of-the-art methods

In this part, we compare QFDet with state-of-the-art multispectral object detection algorithms: CFT (Fang et al., 2021) (without mosaic and perspective data augmentation), TINET (Zhang et al., 2023), ProbEn (Chen et al., 2022b), sensor fusion-based object detection algorithms: HRFuser (Broedermann et al., 2022), baseline detectors customized for RGBT object detection: Faster R-CNN (Ren et al., 2017), Cascade R-CNN (Cai and Vas., 2018), RetinaNet (Lin et al., 2017b), FCOS (Tian et al., 2019b), ATSS (Zhang et al., 2020b), GFL (Li et al., 2020), and state-of-the-art tiny object detection algorithms: FCOS w/ RFLA (Xu et al., 2022b) and QueryDet (Yang et al., 2022). Experiments are conducted on the RGBTDronePerson and VTUAV-det datasets. The results are shown in Tables 2 and 3.

On RGBTDronePerson. In Table 2, “tiny1”, “tiny2”, “tiny3”, and “small” denote absolute object sizes that are in the range of $[0, 8)$, $[9, 12)$, $[12, 20)$, and $[20, 32]$, respectively. QFDet achieves an $\text{mAP}_{50}^{\text{all}}$ of 42.08% and an $\text{mAP}_{50}^{\text{tiny}}$ of 44.04%. Especially, QFDet surpasses the

² Image copyright ownership belongs to School of Information and Communication Engineering, Dalian University of Technology.

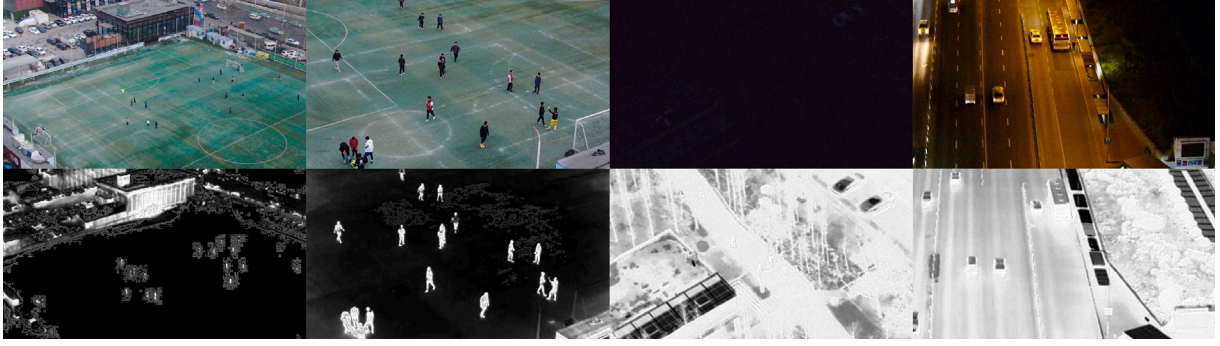


Fig. 9. Example data of VTUAV-det. The first row shows the RGB examples and the second row shows the thermal examples. The object scales and views of VTUAV-det have greater variation. The thermal imaging results are different from that of RGBTDronePerson.

Table 2

Performance comparison with State-of-the-Art methods on RGBTDronePerson. Mark [†] indicates that the method is modified to the version of the RGBT baseline detector. Mark * means that the detection head utilizes P2-P6 feature maps. The bold font demonstrates the best results.

Method	Backbone	mAP ₅₀ ^{all}	mAP ₅₀ ^{tiny}	mAP ₅₀ ^{tiny1}	mAP ₅₀ ^{tiny2}	mAP ₅₀ ^{tiny3}	mAP ₅₀ ^{small}	mAP ₂₅ ^{all}	FPS	Param. (MB)	GFLOPs
ProbEn ^a	ResNet50-FPN	3.64	3.83	0.77	5.41	5.19	0.33	10.73	–	–	–
CFT	CSPDarknet-PANet	22.69	22.83	16.72	18.14	27.52	8.14	37.32	10.8	206.26	112.02
HRFuser	HRFormer-HRFPN	22.23	22.50	0.00	26.73	26.26	23.85	33.24	4.5	55.47	54.17
TINet	ResNet50-FPN	28.30	28.60	0.00	24.12	34.99	34.97	40.34	13.0	100.86	92.80
Faster R-CNN [†]	ResNet50-FPN	28.37	28.48	0.00	21.97	34.87	34.40	38.03	15.9	66.42	62.33
Cascade R-CNN [†]	ResNet50-FPN	31.55	31.99	0.00	29.43	37.77	33.61	42.47	11.8	92.35	76.23
DetectoRS [†]	ResNet50-FPN	32.21	32.65	11.55	30.89	38.40	33.71	43.37	4.6	161.49	74.94
RetinaNet [†]	ResNet50-FPN	22.87	23.34	4.69	13.57	32.66	15.77	38.92	19.4	59.56	49.48
FCOS [†]	ResNet50-FPN	29.89	30.71	9.40	22.73	34.87	26.00	45.21	19.2	78.32	47.21
ATSS [†]	ResNet50-FPN	36.24	37.47	16.92	24.16	43.59	24.62	53.14	19.0	55.31	48.73
GFL [†]	ResNet50-FPN	38.77	40.00	17.86	28.15	46.66	26.75	55.59	18.7	55.45	49.22
FCOS [†] w/ RFLA	ResNet50-FPN	38.20	39.45	25.87	30.93	44.31	25.32	51.41	13.9	78.32	110.62
QueryDet [†]	ResNet50-FPN	37.07	37.75	17.89	24.90	44.05	25.52	55.16	12.8	62.58	121.67
QFDe [*] (Ours)	ResNet50-FPN	42.08	44.04	20.27	30.09	50.36	26.78	57.34	14.2	60.19	81.43
QFDe [*] (Ours)	ResNet50-FPN	46.72	48.75	22.15	37.91	53.71	28.41	61.62	5.7	60.25	242.82

^a The FPS, model parameters, and GFLOPs of ProbEn are not given since the final results are obtained by probabilistic ensembling of the outputs of three separately-trained models.

second competitor GFL[†] by 4.04 points in terms of mAP₅₀^{tiny}. QFDe^{*} achieves the best detection accuracy with an mAP₅₀^{all} of 46.72% and an mAP₂₅^{all} of 61.62% and still keeps a small parameter amount.

Multispectral object detection methods like CFT and TINet take up many model parameters and much computation complexity in the fusion method. For example, CFT utilizes several transformers to realize adaptive fusion. However, these fusion schemes tend to yield unsatisfying results on tiny objects. ProbEn performs poorly on RGBT-DronePerson, which could be that it fuses the results of thermal only, early fusion, and middle fusion models; due to the large position shift between modalities, the early fusion results are quite poor, thereby affecting the final fused results. We have also compared with sensor fusion-based methods: HRFuser. It takes HRFormer (Yuan et al., 2021) (High-Resolution Transformer) as the backbone, which preserves the high resolution of feature maps and thus overcomes the problem of information lost on tiny objects. However, its inference speed notably degrades compared with other methods. Moreover, it is intriguing to see that the two Transformer-based techniques do not show promising performance on RGBTDronePerson. This performance degradation echos the previous study that the global receptive field of the Transformer may not always be beneficial to tiny objects (Carion et al., 2020; Zhu et al., 2021a), indicating the intense challenges brought to Transformer-based fusion detectors in our task. Developed from Faster R-CNN[†], TINet improves the mAP₂₅^{all} by 2 points with its RGBT fusion strategy. However, the contribution made by the fusion strategy is smaller than the detection framework, e.g., Cascade R-CNN. This is because, under the severe challenge of tiny objects, the RGBT fusion improvement alone demonstrates a less significant effect. QFDe^{*} prevails

in six out of seven detection performance metrics except for mAP₅₀^{small}. One possible reason may be that the objects in the small scale interval [20, 32] only account for 2%, leading to the fluctuation.

A qualitative comparison between QFDe^{*}, ATSS[†], and TINet is shown in Fig. 10. The detection performance on tiny objects of QFDe^{*} largely surpasses TINet. Moreover, compared with TINet in the first scene, which use predicted illumination condition to weigh visible and thermal modalities, QFDe^{*} manages to reduce the false positives caused by the heat sinks of the signal tower and shows better robustness. Compared with ATSS[†], QFDe^{*} utilizes RGBT information more effectively. This can be seen in the third scene: QFDe^{*} successfully detects objects hidden in the heated background while the others do not. Tiny object detection methods RFLA and QueryDet greatly elevate the detection performance on tiny1 objects. However, they are not as effective as the QFDe^{*} on tiny2 and tiny3 objects.

On VTUAV-det. QFDe^{*} achieves the mAP of 31.10% and mAP₅₀ of 70.40%. QFDe^{*} elevates the mAP_s by 5.6 points and achieves the highest mAP of 33.30%. In the VTUAV-det testing set, small objects make up 20% of total objects. On mAP_s, QFDe^{*} surpasses the GFL[†] by 3.00 points. On mAP₅₀, QFDe^{*} surpasses the GFL[†] by 3.10 points. The overall performance (mAP) consistently is better than GFL[†] by 1.40 points. On VTUAV-det, which contains a variety of object scales, TINet shows its effect compared with its baseline detector Faster R-CNN[†]. Nevertheless, the improvement is limited. A qualitative comparison between QFDe^{*}, ATSS[†], and TINet is shown in Fig. 11. It can be seen in the first and third scenes, the detection performance on tiny persons of QFDe^{*} is better than ATSS[†] and TINet. By predicting cross-modality

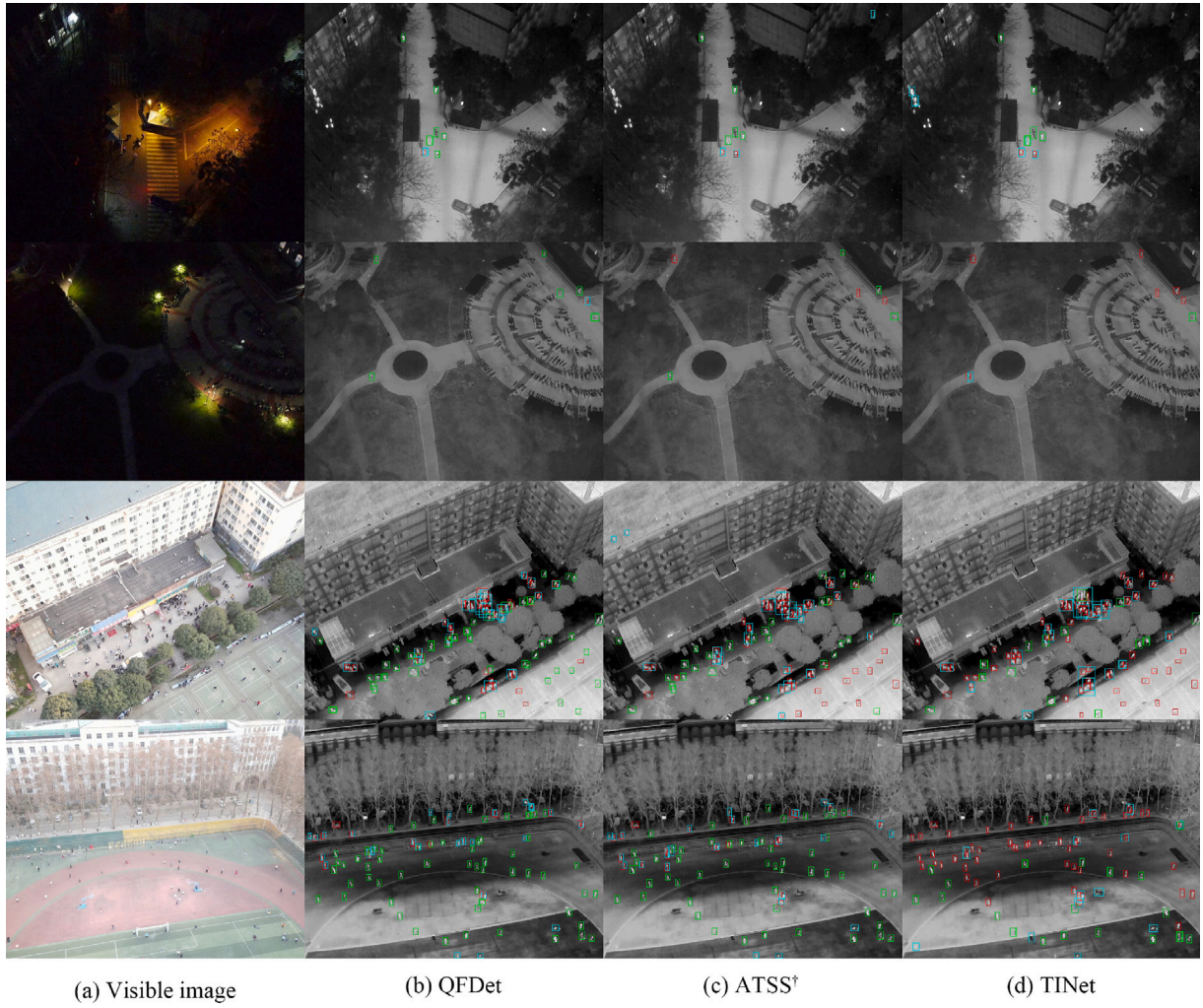


Fig. 10. Visualized detections on RGBTDronePerson comparing with state-of-the-art methods. Green boxes denote true positives, blue boxes denote false positives, and red boxes denote false negatives. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3

Performance comparison with State-of-the-Art methods on VTUAV-det. Mark [†] indicates that the method is modified to the version of the RGBT baseline detector. Mark * means that the detection head utilizes P2-P6 feature maps. The bold font demonstrates the best results.

Method	mAP	mAP ₅₀	mAP ₇₅	mAP _s	mAP _m	mAP _l	FPS
ProbEn	12.70	36.40	5.20	8.50	14.20	14.80	–
CFT	8.73	29.31	2.41	3.96	8.51	23.46	8.3
HRFuser	25.90	55.90	20.10	2.70	27.90	51.90	5.6
TINet	26.80	59.40	20.10	1.20	29.00	53.70	14.5
Faster R-CNN [†]	26.50	58.30	20.50	1.20	28.50	54.50	13.1
Cascade R-CNN [†]	27.50	60.00	22.00	1.80	29.60	55.70	13.0
DetectoRS [†]	29.80	63.20	24.20	3.10	32.30	57.90	4.8
RetinaNet [†]	21.00	50.10	14.80	1.90	21.40	54.20	21.7
FCOS [†]	25.20	60.60	17.20	6.00	25.30	51.80	21.8
ATSS [†]	28.70	66.60	20.70	8.70	28.40	55.60	19.0
GFL [†]	29.70	67.30	22.00	9.50	29.60	56.50	20.1
FCOS* [†] w/ RFLA	28.10	69.70	16.80	15.60	27.80	49.40	15.9
QueryDet [†]	29.50	68.90	20.20	7.80	29.90	53.50	14.6
QFDet (Ours)	31.10	70.40	22.90	12.50	30.40	56.80	15.3
QFDet* (Ours)	33.30	75.50	24.20	18.10	32.40	57.20	9.4

qualities, QFDet adaptively adjusts the contribution of each modality and achieves better multi-modal detection results. The improvement on small objects made by QueryDet is limited when a large scale variation is presented. RFLA greatly elevates the detection performance on small objects. However, the performance on large objects is even worse.

5.3. Ablation studies

Table 4 reports the effect of each proposed module on RGBT-DronePerson. When we adopt QLS in the detection head, the mAP₅₀^{all} and mAP₂₅^{all} are boosted by 3.96 and 4.26 points, respectively. Especially, the detection performance on tiny objects (mAP₅₀^{tiny}) surpasses the baseline,

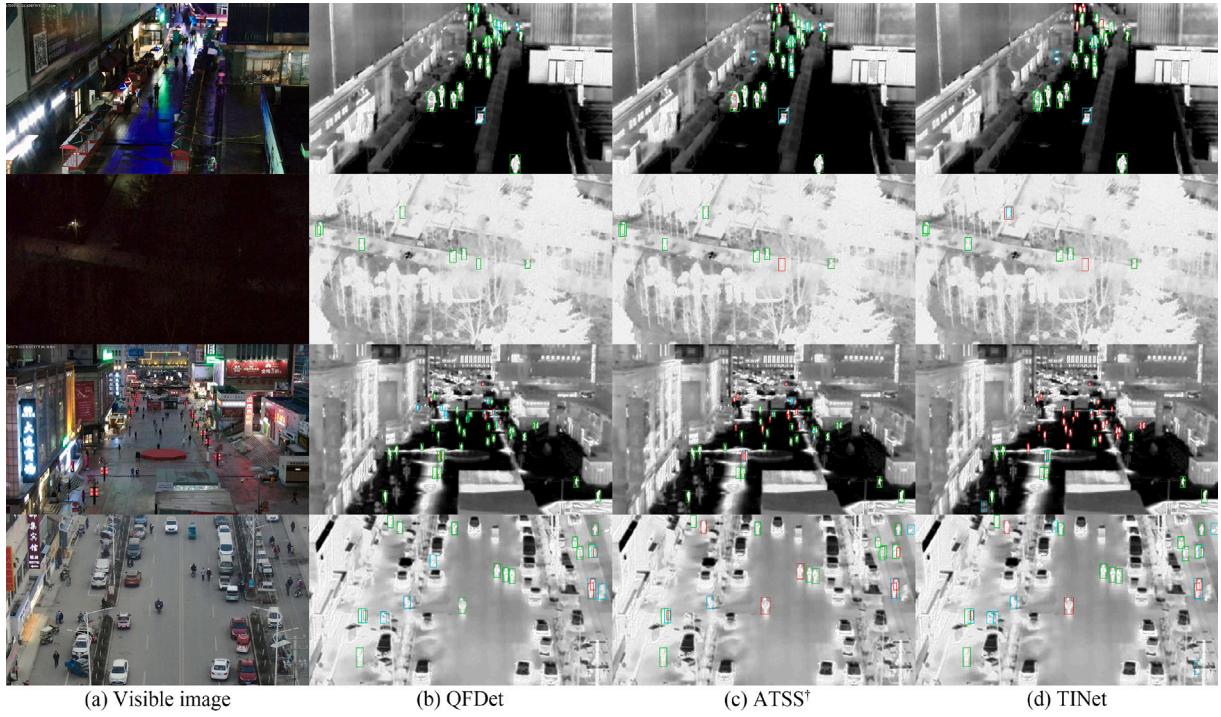


Fig. 11. Visualized detections on VTUAV-det comparing with state-of-the-art methods. Green boxes denote true positives, blue boxes denote false positives, and red boxes denote false negatives. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

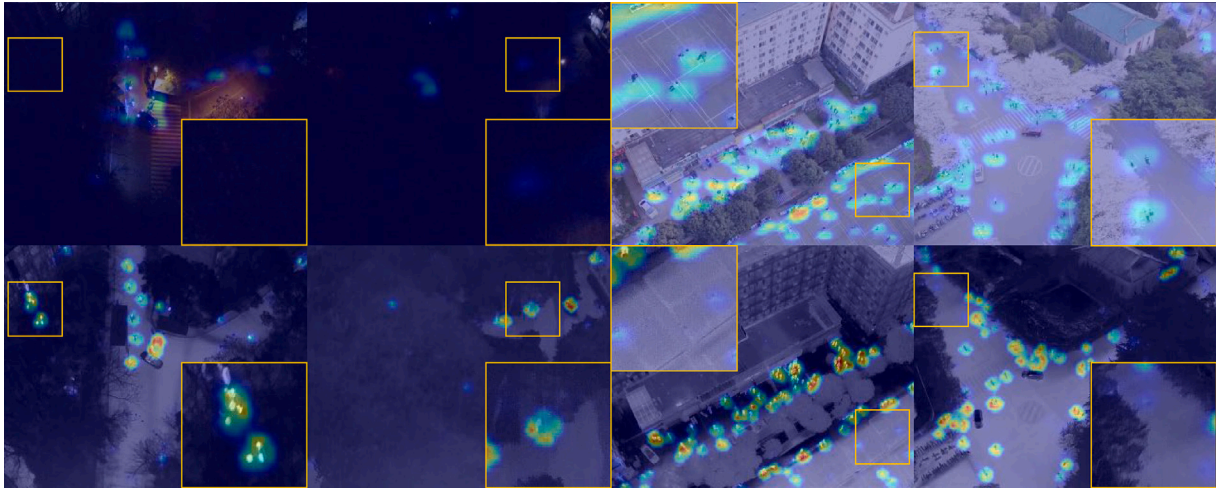


Fig. 12. Examples of predicted QAF maps on RGBTDronePerson. Each column gives an RGBT image pair along with their predicted QAF maps, where the upper and bottom figures show the visible and thermal modality, respectively.

ATSS⁺, by 4.42 points. Moreover, the QCE alone improves the mAP_{50}^{all} and mAP_{25}^{all} by 3.23 and 2.35 points, respectively. We preserve QLS in the PreHead to enable QAF map prediction. With all proposed methods, the detector achieves better detection performance with an mAP_{50}^{all} of 42.08% and an mAP_{50}^{tiny} of 44.04%.

Comparison between different metrics. We validate the effectiveness of the SIWD metric. Tables 5 and 6 show the performance comparison between different metrics on two datasets. On RGBTDronePerson, NWD shows comparable performance with SIWD on tiny objects, while SIWD improves the mAP_{50}^{small} by 3.67 points. The superiority of SIWD is noticeable in the VTUAV-det dataset that consists of various object sizes. Without scale invariance, NWD fails to converge on reasonable and larger objects. In contrast, SIWD improves the learning of all sized objects and thereby elevates detection performance.

Table 4
Ablation study of the proposed methods on RGBTDronePerson.

Method	mAP_{50}^{all}	mAP_{50}^{tiny}	mAP_{50}^{tiny1}	mAP_{50}^{tiny2}	mAP_{50}^{tiny3}	mAP_{25}^{all}
QLS						
QCE	36.24	37.47	16.92	24.16	43.59	53.14
✓	40.20	41.89	18.18	30.83	47.81	57.40
✓	39.47	40.33	11.05	29.86	44.69	55.49
✓	42.08	44.04	20.27	30.09	50.36	57.34

Effectiveness of manhattaness. We validate the effectiveness of the proposed manhattaness on two datasets in Tables 7 and 8. On RGBTDronePerson, manhattaness elevates the mAP_{50}^{all} and mAP_{25}^{all} by 2.45 points and 2.89 points, respectively. Especially for tiny objects,

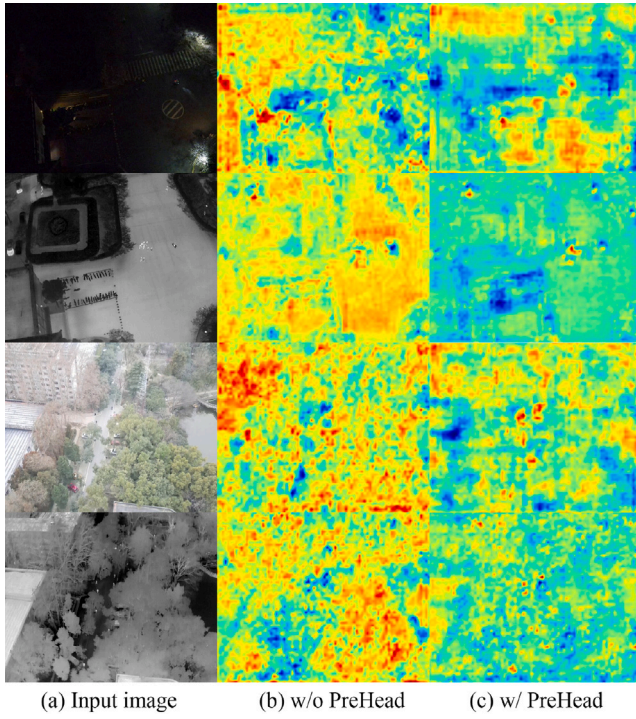


Fig. 13. Examples of how PreHead affects the features. The first column is the input image; the second column is its corresponding backbone-extracted feature map without PreHead; the third column is the feature map with PreHead.

Table 5
Performance comparison between different metrics on RGBTDronePerson.

Method	mAP ₅₀ ^{all}	mAP ₅₀ ^{tiny}	mAP ₅₀ ^{small}	mAP ₂₅ ^{all}
IoU	36.24	37.47	24.62	53.14
NWD	36.58	37.67	27.50	53.37
SIWD	36.97	37.72	31.17	53.31

Table 6
Performance comparison between different metrics on VTUAV-det. Note that the NWD fails to converge on this dataset.

Method	mAP	mAP ₅₀	mAP ₇₅	mAP _s	mAP _m	mAP _l
IoU	28.70	66.60	20.70	8.70	28.40	55.60
SIWD	29.00	67.20	20.90	8.80	28.70	55.70

Table 7
Results of centerness and manhattaness on RGBTDronePerson.

Method	mAP ₅₀ ^{all}	mAP ₅₀ ^{tiny}	mAP ₅₀ ^{small}	mAP ₂₅ ^{all}
centerness	36.24	37.47	24.62	53.14
manhattaness	39.13	40.37	29.58	55.59

Table 8
Results of the centerness and manhattaness on VTUAV-det.

Method	mAP	mAP ₅₀	mAP ₇₅	mAP _s	mAP _m	mAP _l
centerness	28.70	66.60	20.70	8.70	28.40	55.60
manhattaness	29.60	67.90	21.40	9.50	29.60	55.20

the mAP₅₀^{tiny} is higher by 2.90 points. On VTUAV-det, we can see that manhattaness considerably improves the detection performance on small and medium objects.

Effectiveness of QLA. We compare QLA with ATSS (Zhang et al., 2020b) and RKA (Xu et al., 2022a) in Table 9. RKA does not require the positive anchor box center to fall into a GT. For this reason, the centerness will be undefined for anchors whose centers are outside the GT. Therefore we exclude the centerness branch from this experiment

Table 9

Results of different label assignment methods on VTUAV-det. For a fair comparison, these label assignment methods are implemented based on the baseline detector ATSS[†] without the centerness branch.

Method	mAP	mAP ₅₀	mAP ₇₅	mAP _s	mAP _m	mAP _l
ATSS	28.70	66.70	20.70	9.30	28.80	55.50
RKA	26.70	64.60	18.00	10.10	27.00	48.70
QLA	29.30	67.70	20.80	9.50	29.10	54.80

Table 10

Effectiveness of combining the strengths of anchor and prediction box on VTUAV-det. This experiment is conducted based on QFDet.

Method	mAP	mAP ₅₀	mAP ₇₅	mAP _s	mAP _m	mAP _l
Anchor	29.90	68.20	21.30	10.20	29.20	56.50
Prediction	28.10	69.30	16.90	12.70	28.80	50.20
Hybrid	31.10	70.40	22.90	12.50	30.40	56.80

Table 11

Results of QFDet on RGBTDronePerson, without and with the guidance of QAF maps.

QAF map	mAP ₅₀ ^{all}	mAP ₅₀ ^{tiny}	mAP ₅₀ ^{small}	mAP ₂₅ ^{all}
	40.46	42.16	24.17	57.23
✓	42.08	44.04	25.08	57.34

for a fair comparison. Besides, SIWD is adopted for the three methods as metrics. RKA performs well on small objects but the performance degrades on medium and large objects. In comparison, QLA improves performance for various object sizes consistently.

We delve deeper into the effectiveness of the hybrid location quality in QLA. The results presented in Table 10 demonstrate that neither relying solely on prior information (anchors) nor solely on predictions to assess the feature points' location quality leads to the best performance. The underlying reasons can be explained as follows. On the one hand, heuristic and pre-defined anchors cannot fully reflect the feature points' location ability. Given that prior-based assessment works in a prediction-agnostic way, some anchors holding a high prior location score may lie in the background, while some anchors holding a low prior location score may be located on the semantic salient parts. On the other hand, relying entirely on prediction boxes for the label assignment process can have a devastating influence on mAP_l. Due to the instability of predictions at the early training stage, entirely using predictions to assess feature points' quality cannot guarantee a reliable metric. However, in QLA, we combine the strengths of anchors and predictions to select high-quality positive samples. As shown in the last row of Table 10, this hybrid approach effectively elevates the overall performance.

Predicted QAF maps. In Fig. 12, we visualize the predicted QAF maps on RGBTDronePerson testing data. In general, thermal modality shows greater quality response on objects. The first reason is that heated humans are salient in thermal modality. The second reason, in this case, is that thermal modality is aligned with the ground truth annotations. For example, in the second scene, the thermal QAF map responds to the heated human object while in the visible QAF map, it is invisible. Moreover, it is noteworthy that the visible modality information that complements the thermal modality is effectively extracted by the QAF maps. In the zoom-in area of the first scene, the heat sinks of the signal tower is mistakenly predicted as high quality, while with visible modality, the response is suppressed. In the zoom-in area of the third and fourth scenes, human objects are concealed by the background heat in the thermal modality, while in the visible modality, they are clear and distinct. Therefore, false alarms and missed detections can be reduced with QFDet.

We also give quantitative results on the effect of QAF maps in Table 11. With the guidance of QAF maps in the fusion stage, the mAP₅₀^{all} and mAP₅₀^{tiny} are improved by 1.62 and 1.88 points, respectively.

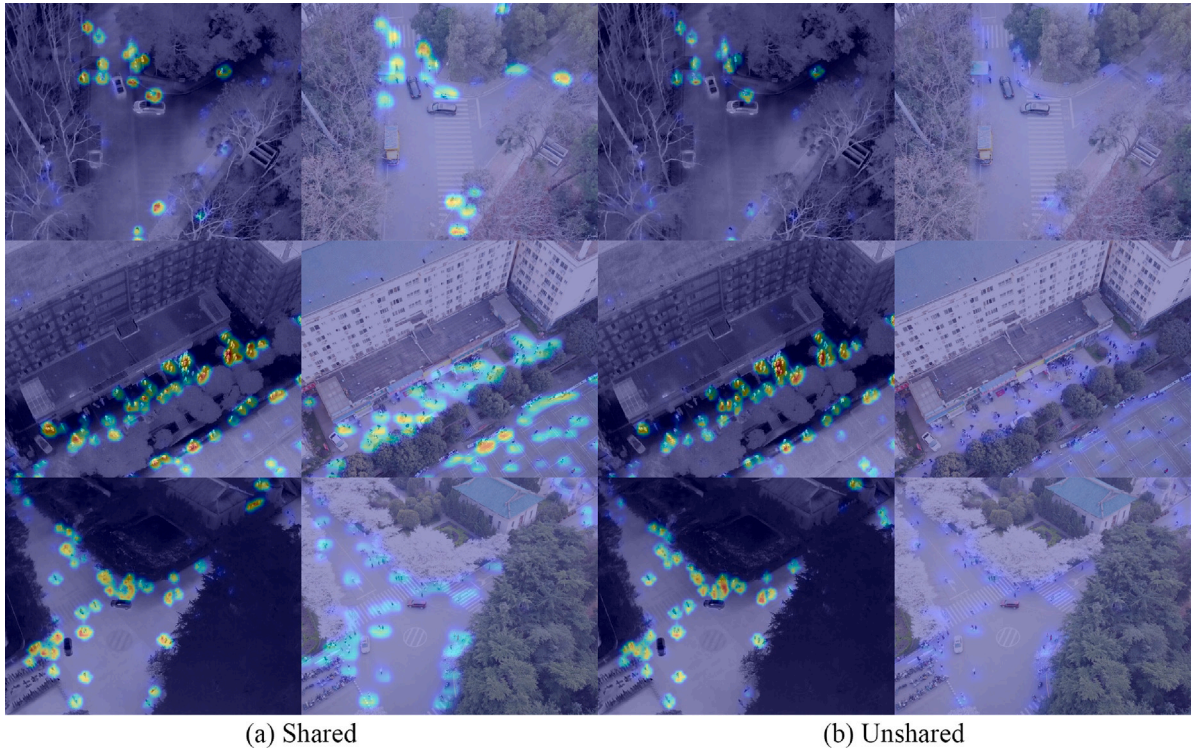


Fig. 14. Comparison between the predicted QAF maps by shared PreHead and unshared PreHead.

Effect of PreHead. Before the features are fused and sent into the final detection head, PreHead is used to predict QAF maps for the visible and thermal features. The quality is defined by localization and classification quality. Therefore, the PreHead also performs the bounding box localization and classification tasks during training. In this way, QFDeT in fact will look twice at the features before yielding final predictions. The first time is to predict with the single modality features and the second time is to predict with the fused features. We have visualized the backbone-extracted visible and thermal feature maps with and without PreHead in Fig. 13. It can be seen that by predicting each modality in advance, the network learns to extract clearer foreground features with less background distraction.

It should be noted that we deploy a shared PreHead for the two modalities. For one thing, this could reduce the model parameters and increase inference speed. For another, given the abundantly shared low-level and high-level information of the thermal and visible modalities, a shared head can produce reasonable predictions for both modalities. We have also implemented an unshared version of PreHead, in which the *conv* layers, *reg*, *cls*, centerness, and quality heads of the thermal and visible modalities are all not shared. Fig. 14 gives the predicted QAF maps of unshared and shared PreHead. It can be seen that in the unshared PreHead, the visible features are predicted with extremely low qualities, while the thermal features are predicted with high qualities in both shared and unshared settings. This is because the ground truth annotations shift away from objects in the visible modality. The separated visible head is unable to learn object localization and classification correctly.

Effectiveness of the pool-upsample operation. We apply max-pooling on the visible features to alleviate the influence brought by the misalignment problem. Then we upsample the pooled feature and fuse it with its original features. Here we dub this operation pool-upsample. In Table 12, we compare QFDeT with and without pool-upsample. It can be seen that the pool-upsample operation improves the mAP_{50}^{all} by 1.68 points and the mAP_{50}^{tiny} by 2.40 points. To demonstrate the effect

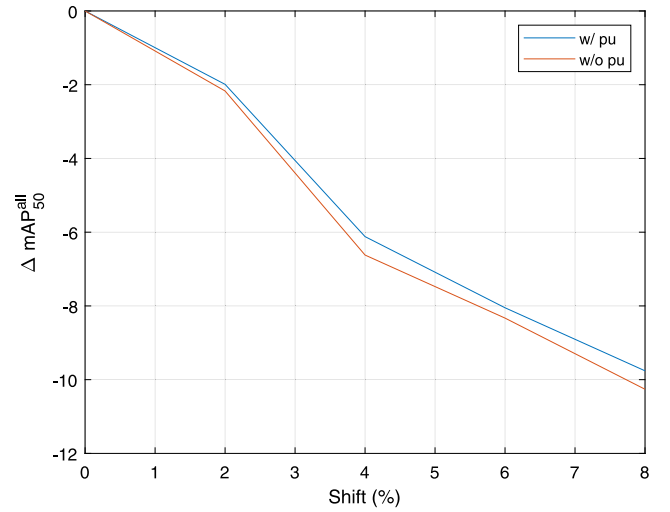


Fig. 15. The detection accuracy degradation amid added shifts of the models with and without pool-upsample. “pu” denotes pool-upsample. The percentage means the max ratio between the shifted pixels between total width/height.

of the approach more clearly and straightforwardly, we randomly add shifts to the visible image in the RGBT image pair and compare the detection performance between models with and without the pool-upsample operation. The results are shown in Fig. 15. It can be seen that with the pool-upsample operation, the resistance of the network toward position shifts is stronger. Moreover, we visualize the visible features before and after the pool-upsample operation in Fig. 16. It can be seen in the first row of feature maps that the pool-upsample

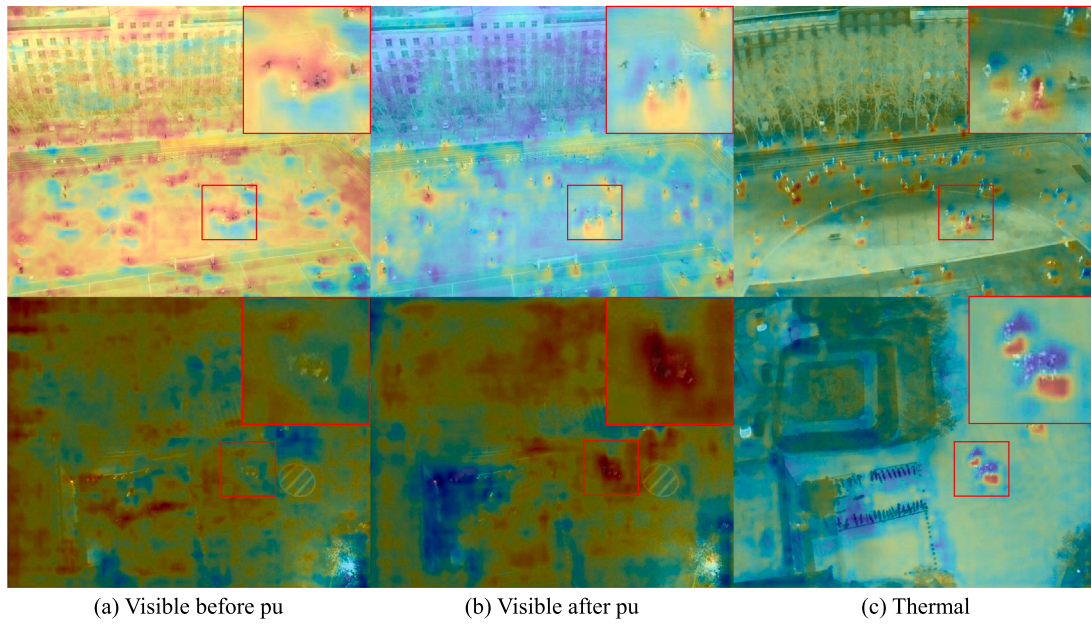


Fig. 16. Feature maps before and after the pool-upsample operation. “pu” denotes the pool-upsample operation.

Table 12

Results of QFDet on RGBTDronePerson, without and with Pool-upsample.

Pool-upsample	mAP ₅₀ ^{all}	mAP ₅₀ ^{tiny}	mAP ₅₀ ^{small}	mAP ₂₅ ^{all}
	40.40	41.64	27.05	57.28
✓	42.08	44.04	25.08	57.34

operation helps discriminate between the objects and the background. And in the second example, the object regions are further addressed.

5.4. Discussion

In this study, we leverage the advantages of drone vision and the fusion of RGBT images to provide time-spatial flexibility for person detection, where we contribute two datasets and a detector called QFDet. Comprehensive statistics have assessed the properties of the data, and extensive experiments have demonstrated QFDet’s efficacy, especially for tiny objects and thermal and illumination imbalance issues. Nevertheless, there are some open problems remaining to be solved, we illustrate these problems from the aspects of data and method.

In terms of data, the person detection task itself limits the diversity of the data, where we only have the “person” category in the proposed datasets. Actually, more tasks can benefit from the time-spatial flexibility of RGBT drone vision, for example, we can divide “person” into more subcategories, for example, “lying person”, “walking person”, and “running person”, so that we can better capture the state of a person. We can also utilize the drone-based RGBT vision data to investigate the habits of wild animals (Benjamin et al., 2018; Tuia et al., 2022) around-the-clock.

In terms of method, we resort to the translation invariance of the max pooling layer to alleviate the position shift problem. However, the solution is somewhat limited regarding two aspects. First of all, the downsampling operation could reduce the position shifts between feature maps, but the information about tiny objects could be lost in the process. Especially in RGBTDronePerson, the position shifts are even greater than the objects themselves. In our pool-upsample operation, we preserve the original features to avoid the loss of tiny object information. Secondly, the method relies on rigid operation. The network still cannot recognize the position shifts between objects

and thereby cannot adjust features adaptively. While other existing methods require ground truth annotations for both modalities, which is labor-intensive. We have considered a possible solution that utilizes the thermal detection head to generate aligned ground truth bounding boxes for the visible modality. To achieve this goal, cross-modality domain adaptation methods (Chen et al., 2018; Vs et al., 2022) and the idea of label noise correction (Liu et al., 2022) are expected to be utilized. Future work includes (i) utilizing a shared detection head to generate aligned ground truth bounding boxes for the visible modality to promote visible feature extraction, (ii) utilizing semantic alignment approaches in multi-modal learning to enhance the RGBT feature representation under the situation of position shifts.

6. Conclusion

In this paper, we construct a benchmark for drone-based RGBT person detection: RGBTDronePerson. The task of drone-based RGBT person detection presents several challenges: tiny object sizes, prominent position shifts between modalities, and modality-space imbalance. Additionally, we annotate another drone-based RGBT person detection dataset VTUAV-det which contains more scale variations than RGBTDronePerson. To tackle these challenges, we propose a QFDet which addresses the imbalance problem by predicting localization and classification quality for each modality. And we design QLS to improve the detection performance on objects of various sizes (especially for tiny objects).

We hope that our open-access datasets will encourage research on the new challenges put forward by the drone-viewed RGBT person detection task. Together with these two datasets, the effectiveness and generalizability of different detectors can be fairly compared. Furthermore, we hope this work could draw effective around-the-clock search and rescue nearer to our reality.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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